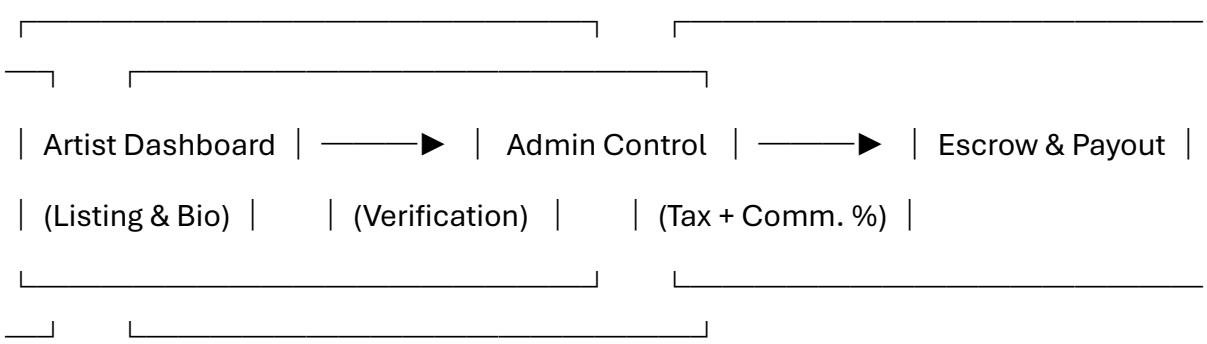

Multi-Vendor Art Marketplace Prototype Workflow



Part 1: The Artist Dashboard (Vendor Portal)

The Artist Dashboard is a restricted control panel accessible only to users with the `role: "artist"`. It empowers artists to manage their public presence, listings, and revenue independently.

1. Profile & Bio Management

* **Artist Bio & City Tag:** The artist creates an authentic public profile. They enter their display name, artist statement/bio, city location (e.g., Kigali, Huye), social links, and profile photo.

* **Payout Account Settings:** The artist connects their mobile money payout details (**MTN MoMo** or **Airtel Money** phone number) or bank account details. This

account is locked once verified and can only be updated with Admin approval for security.

2. Artwork Upload & Inventory Management

* **Add New Piece:** The artist uploads original physical artwork or limited edition prints with specific fields:

* Artwork Title & Detailed Description

* Art Medium (e.g., Oil on Canvas, Acrylic, Digital Print, Sculptures)

* Dimensions & Framing status

* Price in local currency (**RWF**)

* High-Resolution Photos (Front view, detail view, framed view)

* **Status Indicators:**

* `Draft`: Saved locally by the artist, not yet submitted.

* `Pending Review`: Submitted to the Admin; awaiting quality and authenticity approval.

* `Published`: Active in the public gallery and available for purchase.

* `Sold`: Artwork purchased by a customer (inventory automatically drops to `0`).

3. Sales & Order Management

* **Order Notifications:** When a customer orders a piece, the artist receives an SMS and email notification containing order details and shipping instructions.

* **Dispatch & Packaging:** The artist updates the order status to `Ready for Pickup` or `Shipped` and uploads a photo of the packaged artwork along with a tracking/courier receipt.

4. Financial History & Ledger

* **Real-time Balance Overview:** Displays **Total Gross Sales**, **Platform Commissions Deducted**, **Taxes Withheld**, and **Net Available Payout**.

* **Withdrawal Requests / Payment Logs:** A transparent statement showing every completed sale, showing exactly how much money was received, how much was deducted, and the transaction hash/reference when funds land in their Mobile Money wallet.

Part 2: Admin Control & Step-by-Step Monitoring

The Admin Panel provides complete supervisory control, ensuring quality assurance, preventing fraud, and enforcing financial compliance.

1. Artist Onboarding Verification

* **Application Review:** When an artist registers at `/join-as-artist`, their application enters the `/admin/artists` queue in `Pending` status.

* **Identity Check:** Admin inspects portfolio links, phone numbers, and identity documents.

* **Action:** Admin clicks **Approve** (enabling the user's `artist` dashboard) or **Reject** (with feedback sent via email).

2. Artwork Moderation Before Listing

* Every newly created piece enters the `/admin/catalog` approval queue.

* Admin verifies photo quality, pricing realism, and content guidelines before clicking **Publish**. This guarantees that low-quality or fake listings never reach the public gallery.

3. End-to-End Delivery Monitoring (Escrow Workflow)

To protect both the buyer and the platform, the platform uses an **Escrow Verification Model**:

```
[Customer Pays] ——▶ [Escrow Hold] ——▶ [Delivery Confirmed] ——▶ [Deduct Tax/Comm] ——▶ [Release to Artist]
```

1. **Step 1: Payment Lock:** When a customer pays, money enters the primary **Platform Escrow Account**. The artwork status flips to `Sold`.
2. **Step 2: Fulfillment Tracking:** Admin monitors shipment status on `/admin/orders`. The artist drops off the piece with the platform's courier or logistics partner.
3. **Step 3: Delivery Confirmation:** The customer receives the item and signs for delivery (or confirms via a digital OTP code / website click).
4. **Step 4: Dispute Window:** A 24–48 hour inspection period opens for the buyer to confirm the piece matches the photos and is undamaged.

Part 3: Financial Calculations (Tax, Commission, and Payout)

Once delivery is successfully verified, the system automatically executes the financial deduction formula before initiating the transfer (`cashout`) to the artist's MTN/Airtel wallet.

1. Deduction Formula & Financial Logic

The system processes three components:

1. **Gross Sale Price (P):** The price listed by the artist.

2. **Platform Commission Percentage (C):** Platform service fee (e.g., 15%).

3. **Withholding Tax (T):** Local government withholding tax on platform earnings or digital service transactions (e.g., 15% withholding or standard VAT rules depending on local tax registration).

$$\text{Platform Commission Fee} = P \times C$$

$$\text{Withholding Tax Amount} = P \times T$$

$$\text{Artist Net Payout} = P - (\text{Platform Commission Fee} + \text{Withholding Tax Amount})$$

---

2. Numerical Example Breakdown (RWF)

Suppose an artist sells a painting for **100,000 RWF**:

* **Platform Commission Rate:** 15%

* **Withholding Tax Rate:** 15%

```

Gross Sale Price:                    100,000 RWF

_____

(-) Platform Commission (15%):          -15,000 RWF  (Platform Earnings)

(-) Tax Withheld (15%):                -15,000 RWF  (Remitted to Tax Authority)

_____

(=) Net Amount Payable to Artist:          70,000 RWF  (Disbursed via MoMo API)

```

---

3. Automated Payout Execution

* **Ledger Logging:** The system logs this exact breakdown into the `ArtistLedger` database table with `payout_status: "approved"`.

* **API Trigger:** The server calls the **Paypack** or **Flutterwave** `cashout/transfer` API endpoint.

* **Instant Disburse:** The **70,000 RWF** is transferred directly to the artist's MTN or Airtel Mobile Money account.

* **Receipt Generation:** An itemized breakdown (PDF/email statement) is sent to both the artist and the admin for tax compliance auditing.

Summary Table: Admin vs. Artist Control Points

| Stage | Artist Role | Admin Action / Control | System Automated Logic |

| --- | --- | --- | --- |

| **1. Registration** | Fills profile & MoMo payout info. | Verifies identity; approves account. | Locks account settings once verified. |

| **2. Listing** | Uploads image, dimensions, & price. | Reviews image quality & approves listing. | Publishes art to public `/gallery`. |

| **3. Purchase** | Receives order notification & packages art. | Monitors order in `/admin/orders`. | Collects customer funds into Escrow. |

| **4. Delivery** | Hands artwork to delivery courier. | Verifies proof-of-delivery from buyer/courier. | Triggers 24-hour dispute countdown timer. |

| **5. Settlement** | Receives net payment notification on MoMo. | Audits monthly tax & platform earnings. | Deducts tax + commission; disburses net balance. |